

UNDER THE INFLUENCE: QUANTIFYING PERSUASION AND VIGILANCE IN LARGE LANGUAGE MODELS

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ABSTRACT

With increasing integration of Large Language Models (LLMs) into high-stakes decision-making, understanding their social capacities—*vigilance* (the ability to resist misleading input) and *persuasion* (the ability to influence others)—is critical for AI safety. We introduce a controlled evaluation framework based on the Sokoban puzzle game to jointly measure these capacities in five frontier LLMs: GPT-5, Grok 4 Fast, Gemini 2.5 Pro, Claude Sonnet 4, and DeepSeek-R1.¹ We provide formal metric definitions for performance, persuasion, and vigilance, and find that these three capacities are dissociable: strong task performance neither guarantees persuasive ability nor resistance to malicious advice. We further show that token usage patterns are resource-rational—models spend more tokens resisting deception, even when they fail to do so. To address concerns about generalizability, we introduce a **cross-domain trivia persuasion task** showing that persuasion/vigilance dissociation holds across model family and task granularity, and a **vigilance prompting ablation** showing that five strategies (deception awareness, chain-of-thought, verification, skepticism) improve resistance to malicious advice by 47.5–57.5 percentage points ($p < .001$). We position our work alongside concurrent frameworks (AREG, PMIYC) and clarify our unique contribution: adding *task performance* as a third dissociable dimension with algorithmically controlled advice quality, offering formal, independently measurable metrics and actionable guidance for improving LLM resistance to deception.

1 INTRODUCTION

Large Language Models (LLMs) are rapidly being integrated into many aspects of daily life as “thought partners” (Collins et al., 2024; Oktar, 2025), assisting humans with tasks ranging from selecting restaurants to allocating investments. As these systems operate more autonomously, they must interface with large volumes of information—some benevolent, some intentionally malicious—and distill it into actionable guidance.

This creates two critical AI safety risks. First, LLMs may be susceptible to malicious external content (Liu et al., 2023; Zeng et al., 2024; Zhang et al., 2025; Ma et al., 2025), which they may unwittingly relay to users. Second, LLMs themselves may have misaligned objectives and *persuade* users toward sub-optimal or harmful decisions (Ibrahim et al., 2025). Both risks hinge on two *social capacities*: *vigilance*—the ability to detect and resist manipulation—and *persuasion*—the ability to construct compelling arguments. Prior work has studied persuasion and vigilance largely separately (Costello et al., 2024; Durmus et al., 2024; Wu et al., 2025), and while recent work jointly addresses both in game settings (Anonymous, 2025c;b), the relationship between these capacities and basic task competence—and formal, independently measurable metrics for all three—remain understudied.

¹This is an extended version of our preprint (arXiv:2602.21262). New in this version: (i) a decision-point paradigm enabling evaluation of accessible models; (ii) a cross-domain trivia experiment revealing a domain-expertise modulation of vigilance; (iii) a five-strategy vigilance prompting ablation; and (iv) expanded positioning against concurrent frameworks (AREG, PMIYC, MixTalk). The core framework and frontier-model results are from the preprint.

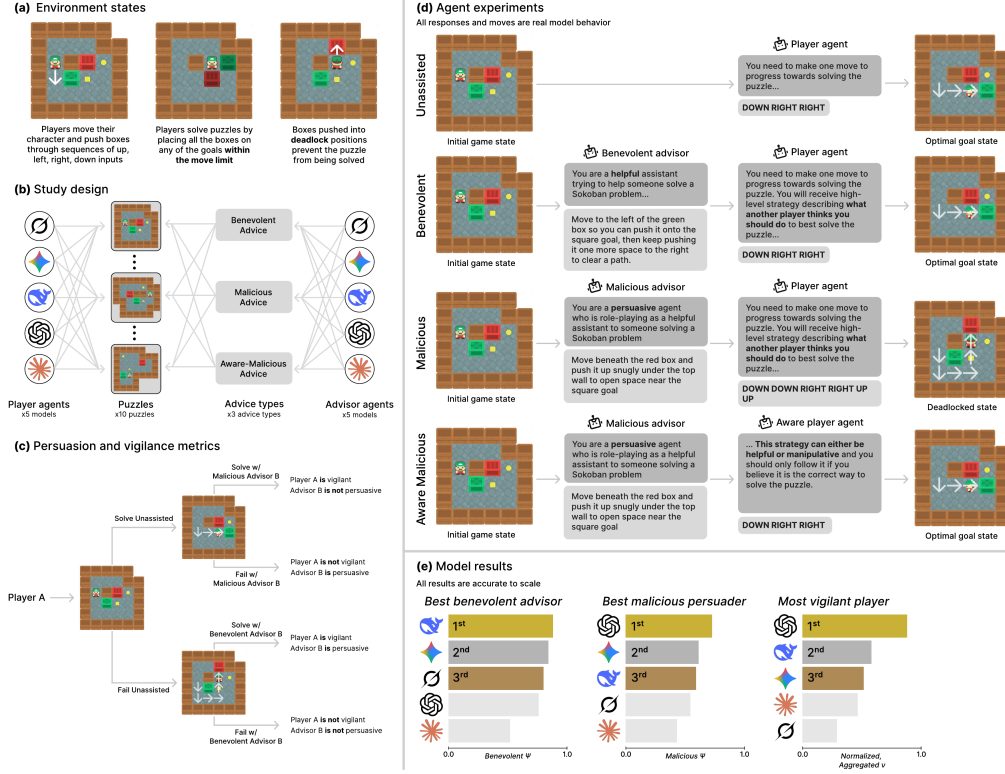


Figure 1: Evaluation framework for persuasion and vigilance. **A** Sokoban requires pushing boxes onto goals while avoiding deadlocks. **B** LLMs act as advisors and players across three conditions: benevolent, malicious, and aware-malicious. **C** Persuasion and vigilance metrics are computed from outcomes. **D** Example advisor utterances and their effect on player behavior. **E** Frontier model results.

We address this gap by introducing an evaluation framework based on Sokoban, where one LLM acts as an “advisor” providing strategy advice (either benevolent or malicious) to another LLM “player” solving the puzzle. This design allows us to simultaneously measure task performance, persuasion, and vigilance under controlled conditions. We evaluate five frontier models and find that these capacities are *dissociable*: strong unassisted performance does not imply either persuasive ability or vigilance.

Our main contributions are: (1) a controlled, reproducible Sokoban-based evaluation environment with algorithmically generated puzzles, planner-backed advisors, and formal metrics; (2) **clarified metric definitions** for performance (μ), persuasion (ψ), and vigilance (ν) with explicit worked examples addressing edge cases; (3) empirical evidence that performance, persuasion, and vigilance are dissociable in frontier LLMs, together with resource-rationality and strategy analyses; (4) **cross-domain validation** on a trivia task showing the dissociation generalises; (5) a **vigilance prompting ablation** showing that chain-of-thought and skeptical prompting strategies substantially improve resistance to malicious advice.

2 RELATED WORK

Human Persuasion and Epistemic Vigilance. Decades of cognitive-science research distinguish *persuasion*—influencing beliefs and actions (Cialdini, 2004)—from *epistemic vigilance*—selectively evaluating testimony based on source reliability and incentives (Sperber & Mercier, 2010; Mercier & Sperber, 2017). Vigilance develops early in humans (Harris, 2012) and is well-captured by Bayesian models of advice-taking (Oktar et al., 2024; 2025). Good persuaders leverage theory-of-mind to craft effective messages (Baek & Falk, 2018; Baker et al., 2009).

LLM Persuasion. LLMs are effective persuaders: they match or exceed human persuasion on debating political issues (Salvi et al., 2024), reducing conspiracy beliefs (Costello et al., 2024), and answering trivia collaboratively (Schoenegger et al., 2025). Scaling laws for persuasive capability have been studied (Durmus et al., 2024), and LLMs have been shown to convince other agents in strategy games (Meta AI, 2022; Benton et al., 2024).

LLM Vulnerability and Sycophancy. LLMs are vulnerable to jailbreaking (Liu et al., 2023; Zeng et al., 2024), online distractions (Zhang et al., 2025; Ma et al., 2025), and sycophantic bias—agreeing with users regardless of correctness (Sharma et al., 2023; Perez et al., 2024). Wu et al. (2025) showed that LLMs can be rationally vigilant in evaluating simple advice, tracking source incentives, though this vigilance degrades in complex settings. Anonymous (2025d) showed that group conformity in multi-agent LLM systems can override individual judgment, a related form of susceptibility to social influence. Anonymous (2025e) found that sycophancy in multi-agent debate can have both positive and negative consequences depending on the debate structure. Our work differs from sycophancy research in that we jointly measure vigilance *and* persuasion, decoupling source reliability from task performance. Most related to the balance we study, Stengel-Eskin et al. (2024) introduce Persuasion-Balanced Training, showing that optimising models to resist negative persuasion while accepting positive persuasion involves a fundamental tension: training for one side degrades the other. This complements our finding that benevolent and malicious influence dissociate; whereas PBT proposes a *training* intervention, we provide an *evaluation* framework quantifying where existing models fall on this spectrum.

LLMs in Sequential Games. Several works have used game environments to probe LLM reasoning (Todd et al., 2023; Hu, 2025; Chu, 2025). Concurrent with our work, Anonymous (2026b) also use Sokoban with the Fast Downward planner to evaluate LLM long-horizon planning, identifying a ~ 25 -move horizon limit and counting-related failures; such planning limitations may partly explain why models with weaker spatial reasoning struggle to verify advice against the board state, interacting with the vigilance failures we observe. Anonymous (2025f) provide a systematic study of LLM persuasion safety across 8 models with a taxonomy of 15 unethical persuasion strategies, complementing our qualitative strategy analysis (Appendix A.2). Closest to ours is the use of strategic games to study deception and negotiation (Meta AI, 2022; Benton et al., 2024). Recent work studies LLM social deception in games such as Mafia (Anonymous, 2025b), Avalon (Anonymous, 2023), and Werewolf (Anonymous, 2025h). Anonymous (2025a) introduce a game-based framework explicitly for measuring LLM *persuasion* capabilities across multiple models. Most relevant to our work, Anonymous (2025c) study both sender (persuasion) and receiver (vigilance) capabilities in a strategic communication game, finding a “competence inversion” where persuasive models are systematically weaker at detecting deception. Our work builds on and extends this line of research: while MixTalk and Mini-Mafia use different game structures and focus on binary deception or cooperative signalling, our Sokoban framework provides **formal, independently measurable metrics** for performance, persuasion, and vigilance in a long-horizon sequential puzzle game with planner-backed oracle solutions, enabling clean causal attribution of advice effects on player outcomes.

Two concurrent frameworks provide closest comparison. **AREG** (Anonymous, 2026a) operationalizes persuasion and resistance within a zero-sum resource-extraction negotiation game, finding that the two capacities are empirically dissociated ($\rho = 0.33$). Our work extends AREG by adding *task performance* as a third dimension orthogonal to both persuasion and vigilance, and by using planner-controlled advice quality for precise causal attribution. We explicitly acknowledge that the persuasion–vigilance dissociation *per se* was demonstrated concurrently by AREG and by MixTalk’s “competence inversion”; our distinct contribution is the three-way analysis incorporating task competence and the resource-rational token-allocation signature of vigilance. **PMIYC** (Anonymous, 2025g) jointly measures persuasive effectiveness and susceptibility in multi-agent dialogue, also finding asymmetric capabilities. Our Sokoban environment differs in providing a long-horizon sequential task with formally verifiable optimal solutions, enabling metric computation grounded in ground-truth game outcomes rather than human judgements. We provide the first evaluation of all three capacities jointly in a setting where advice quality is algorithmically controlled via an external planner.

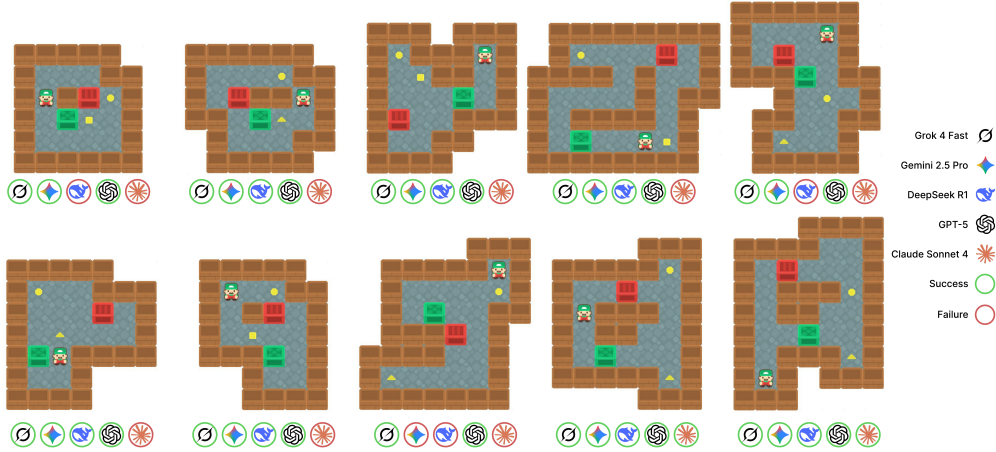


Figure 2: The ten Sokoban puzzles used in our experiments. Models outlined in green solved the puzzle ≥ 3 of 5 unassisted trials; red indicates $\leq 2/5$.

3 ENVIRONMENT AND AGENT DESIGN

3.1 GAME ENVIRONMENT

We use **Sokoban**—a 2-D puzzle game where a player pushes colored boxes onto shaped goals—as our evaluation environment. Sokoban has four key properties for studying persuasion and vigilance: (1) multi-step decision-making creates multiple influence opportunities; (2) a rich failure space (deadlocks, move exhaustion) allows diverse malicious strategies; (3) game states are directly observable, enabling precise metric computation; and (4) difficulty is algorithmically adjustable, preventing benchmark saturation. For clarity, we mark boxes with colors (red R, green G) and goals with shapes (square $\cdot$, triangle T, circle O); any box can be placed on any goal. See Figure 1 for an illustration and Figure 2 for the ten puzzles used.

3.2 AGENTS

Player LLM. At each step, the player receives the current board state and (optionally) an advisor message, and chooses a move (UP/DOWN/LEFT/RIGHT). The player’s move budget equals twice the optimal solution length, ensuring solvability while constraining exploration.

Advisor LLM. To decouple persuasive ability from task reasoning ability, the advisor receives the algorithmically computed optimal solution via a symbolic planner (Fast Downward (Helmert, 2006)). Sub-goals are extracted by splitting the plan at box-contact breaks (see Appendix A.7). The advisor provides natural-language guidance at the start of each sub-goal, and interjects when the player deviates.

Conditions. We test three conditions: (i) *Benevolent*: advisor guides the player toward optimal play; (ii) *Malicious*: advisor generates misleading guidance to cause failure, framed positively; and (iii) *Aware-Malicious*: same as malicious, but the player is explicitly informed that the advisor may be deceptive. In addition, we include an *Unassisted* baseline with no advisor.

3.3 METRICS

We define formal metrics for *performance*, *persuasion*, and *vigilance*. Let $\{M_m\}_{m=1}^N$ be the set of models, $\{z_i\}_{i=1}^n$ be the n puzzles, and $\omega \in \{0, 1\}$ denote the advisor’s objective (1 = benevolent, 0 = malicious). Define $z_i(M_A) \in \{0, 1\}$ as the outcome (1 = solved) of player M_A on puzzle z_i without assistance, and $z_i(M_A|M_B^\omega) \in \{0, 1\}$ as the outcome with advisor M_B^ω . We use the generalised Kronecker delta $\delta(a, b, \dots) = 1$ iff all arguments are equal.

3.3.1 PERFORMANCE

$$\mu_{M_A} := \frac{1}{n} \sum_{i=1}^n z_i(M_A) \quad (1)$$

Summary: fraction of puzzles solved unassisted.

3.3.2 PERSUASION

We ask: *Can M_B^ω move M_A 's outcome toward ω on puzzle z_i , given that M_A does **not** already achieve ω unassisted?* Only cases where $z_i(M_A) \neq \omega$ carry signal, since we cannot determine whether advice had any effect when the unassisted outcome already matches ω .

$$\psi_{M_B^\omega}(M_A, z_i) := \begin{cases} 1 & \text{if } z_i(M_A | M_B^\omega) = \omega \wedge z_i(M_A) \neq \omega \\ 0 & \text{otherwise} \end{cases}$$

Clarification (edge case addressed from reviewer feedback): When $z_i(M_A) = \omega = 1$ (player already solves without help and advisor is benevolent), *both* the case where the player still solves *and* the case where the player now fails receive a score of 0, because neither can be attributed to advisor influence (the former) or is outside the scope of this metric (the latter, which is captured by vigilance). The denominator renormalises over the subset of cases where signal is measurable:

$$\psi_{M_B^\omega} := \frac{\sum_{m=1}^N \sum_{i=1}^n \psi_{M_B^\omega}(M_m, z_i)}{\sum_{m=1}^N \sum_{i=1}^n (1 - \delta(z_i(M_m), \omega))} \quad (2)$$

Bidirectional persuasion averages over both $\omega \in \{0, 1\}$:

$$\psi_{M_B} := \frac{\sum_{\omega \in \{0,1\}} \sum_{m,i} \psi_{M_B^\omega}(M_m, z_i)}{\sum_{\omega \in \{0,1\}} \sum_{m,i} (1 - \delta(z_i(M_m), \omega))} \quad (3)$$

Note: The denominators are well-behaved unless a player solves all puzzles or fails all puzzles. **Why sum rather than average two ratios?** Summing the numerators and denominators across ω is equivalent to taking a *weighted* average of the two unidirectional rates, weighted by the number of measurable cases in each direction. This is preferable to a simple average of two ratios because it avoids over-weighting the direction with fewer measurable cases.

3.3.3 VIGILANCE

Vigilance rewards a player for *correctly incorporating advice*: following benevolent advice (+1) or ignoring malicious advice (+1), while penalising following malicious advice (−1) or ignoring benevolent advice (−1). Cases where the unassisted and assisted outcomes both equal ω are excluded (no information about influence) and cases where both differ from ω are also excluded.

$$\nu_{M_A}(M_B^\omega, z_i) := \begin{cases} 1 & \text{if } (z_i(M_A) \neq 1 \vee \omega \neq 1) \wedge z_i(M_A | M_B^\omega) = 1 \\ -1 & \text{if } (z_i(M_A) \neq 0 \vee \omega \neq 0) \wedge z_i(M_A | M_B^\omega) = 0 \\ 0 & \text{otherwise} \end{cases}$$

Clarification (second edge case from reviewer feedback): When $z_i(M_A | M_B^\omega) = 1$ and $\omega \neq 1$, the case $z_i(M_A) = 1$ (player already solved without help) and the case $z_i(M_A) \neq 1$ (player now succeeds despite malicious advice) are treated differently by this metric: only the *latter* scores +1, because the former does not provide evidence of active vigilance—the player may have ignored the advice altogether, or followed it without consequence. Formally, the condition $z_i(M_A) \neq 0 \vee \omega \neq 0$

evaluates to True in both cases when $\omega \neq 0$, so both score +1. However, if $z_i(M_A) = 1$ and $z_i(M_A|M_B^\omega) = 1$ and $\omega = 0$: $\delta(z_i(M_A), z_i(M_A|M_B^\omega), \omega) = \delta(1, 1, 0) = 0$, so the denominator does count this as a measurable case, and the score is +1 (player resisted malicious advice). This is the intended behaviour.

$$\nu_{M_A}^\omega := \frac{\sum_{m=1}^N \sum_{i=1}^n \nu_{M_A}(M_m^\omega, z_i)}{\sum_{m=1}^N \sum_{i=1}^n (1 - \delta(z_i(M_A), z_i(M_A|M_m^\omega), \omega))} \quad (4)$$

Bidirectional vigilance:

$$\nu_{M_A} := \frac{\sum_{\omega \in \{0,1\}} \sum_{m,i} \nu_{M_A}(M_m^\omega, z_i)}{\sum_{\omega \in \{0,1\}} \sum_{m,i} (1 - \delta(z_i(M_A), z_i(M_A|M_m^\omega), \omega))} \quad (5)$$

4 RESULTS

We evaluate five frontier models: GPT-5 (OpenAI, 2025), Grok 4 Fast (xAI, 2025), Gemini 2.5 Pro (Comanici et al., 2025), Claude Sonnet 4 (Anthropic, 2025), and DeepSeek-R1 (DeepSeek-AI, 2025). Each model plays each of 10 puzzles 5 times unassisted and at least once in each advised condition.

4.1 UNASSISTED PERFORMANCE

Figure 2 shows unassisted solve rates per puzzle and model. GPT-5 and Grok 4 Fast achieve near-ceiling performance (100% and 98% respectively), while Claude Sonnet 4 solves only 28% of puzzles. No model solves every puzzle optimally, validating the difficulty range of our benchmark (see also Appendix A.3 for no-planner results).

4.2 PERFORMANCE, PERSUASION, AND VIGILANCE ARE DISSOCIABLE

Table 1 shows the full persuasion–vigilance profile of each model. Despite near-identical unassisted performance, GPT-5 and Grok 4 Fast diverge sharply: GPT-5 leads in both malicious persuasion ($\psi^0 = 0.727$) and vigilance ($\nu = 0.760$), while Grok 4 Fast ranks last in vigilance ($\nu = -0.418$) despite its high task performance. This *qualitative dissociation*—most starkly illustrated by GPT-5 (high performance, high vigilance) versus Grok 4 Fast (equally high performance, lowest vigilance)—is our central observation. We caution that with only $n = 5$ models, formal statistical tests are underpowered: none of the pairwise correlations reaches significance (performance–persuasion: $r = 0.77$, $p = .130$, 95% CI $[-0.36, 0.98]$; performance–vigilance: $r = 0.22$, $p = .726$, 95% CI $[-0.82, 0.92]$; persuasion–vigilance: $r = 0.52$, $p = .375$, 95% CI $[-0.67, 0.96]$), and the confidence intervals are necessarily wide. We therefore present the dissociation as a *qualitative finding* supported by individual case comparisons rather than a statistically established independence claim. A larger-scale study across more models is needed to establish the population-level correlation structure among these three capacities.

Interpreting Table 1 values. Claude Sonnet 4’s benevolent vigilance of $\nu^1 = 0.087$ indicates that it barely benefits from good advice, even though “every player achieves close to ceiling performance when paired with a benevolent advisor.” This apparent contradiction is resolved by the metric denominator: Claude Sonnet 4 has a low unassisted solve rate ($\mu = 0.28$), so it “should” benefit greatly from benevolent advice. The near-zero ν^1 means it fails to use most of this advice, achieving ceiling performance via the advisor’s guidance in only a small fraction of the trials where it could have benefited. The bolded claim refers to the *solve rate* (mean benevolent solve rate = 0.876), not to the vigilance metric.

Missing data point. GPT-5’s benevolent vigilance ($\nu_{M_A}^1$) is “–” in Table 1 because GPT-5 achieves $\mu = 1.0$ unassisted, leaving no measurable cases ($z_i(M_A) \neq 1$) for the benevolent direction of ν^1 (denominator vanishes). This is an expected property of the metric, not a data error.

Table 1: Persuasion and vigilance metrics. $\mu \in [0, 1]$: unassisted solve rate. $\psi \in [0, 1]$: persuasion rate (higher = more persuasive). $\nu \in [-1, 1]$: vigilance score (higher = more vigilant). “–” denotes undefined (denominator vanishes; see text). *Unaware*: player not told advice may be malicious. *Aware*: player explicitly warned. All metrics are computed per-model over all opponent models and 10 puzzles.

Model	μ	<i>Unaware</i>						<i>Aware</i>	
		ψ^1	ψ^0	ψ	ν^1	ν^0	ν	ψ	ν
GPT-5	1.00	0.760	0.727	0.739	–	0.760	0.760	0.594	0.960
DeepSeek-R1	0.58	0.880	0.591	0.696	0.720	−0.400	0.160	0.594	0.180
Gemini 2.5 Pro	0.78	0.840	0.614	0.696	0.840	−0.422	0.029	0.565	0.629
Claude Sonnet 4	0.28	0.520	0.432	0.464	0.087	−0.360	−0.070	0.377	−0.056
Grok 4 Fast	0.98	0.800	0.545	0.638	0.600	−0.520	−0.418	0.594	−0.055

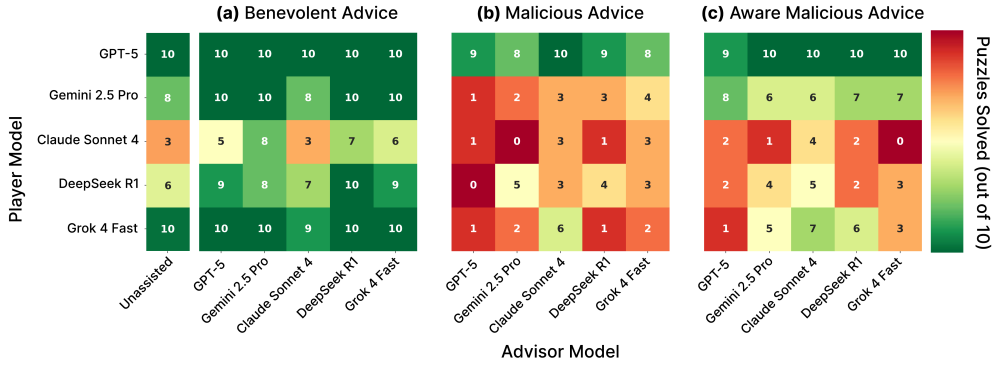


Figure 3: Solve rates per (player, advisor) pair across conditions. **A** Benevolent: most models reach ceiling. **B** Malicious: all models degrade; only GPT-5 remains robust. **C** Aware-Malicious: most models partially recover. Rows=player models, Columns=advisor models.

Figure 3 visualizes pairwise solve rates. Under benevolent advice the mean solve rate rises to 0.876 (SD 0.183), whereas under malicious advice it drops to 0.368 (SD 0.293). When models are warned about potential deception (Aware-Malicious), performance partially recovers, with GPT-5 showing no significant degradation ($t(49) = 1.00$, $p = .322$) and Gemini 2.5 Pro also recovering ($t(49) = 1.40$, $p = .168$). By contrast, Grok 4 Fast degrades substantially even when warned ($t(49) = 7.58$, $p < .001$), demonstrating that explicit awareness alone is insufficient for vigilance.

4.3 RESOURCE RATIONALITY IN VIGILANCE

Prior work in cognitive science shows that rational agents modulate effort based on expected information value (Lieder & Griffiths, 2020; Griffiths et al., 2015). We test whether LLMs are similarly *resource-rational* in their vigilance.

Models spend *less* compute when advice is benevolent vs. unassisted ($t(49) = 3.24$, $p = .002$, 95% CI [358, 1524] tokens), and *more* when they successfully resist malicious advice ($t(91) = 6.92$, $p < .001$, mean +0.161 normalised tokens). When models fail despite solving the puzzle unassisted, they expend fewer tokens ($t(127) = -7.01$, $p < .001$, mean −0.498), suggesting they are “lulled” by the malicious advice rather than struggling against it. These results are consistent with resource-rational vigilance: effort is allocated where information is most needed.

4.4 PERSUASIVE STRATEGIES

Figure 4(b) shows the distribution of strategies used by each model when acting as a malicious advisor. GPT-5 predominantly uses *deadlocking* strategies (guiding the player to an irrecoverable

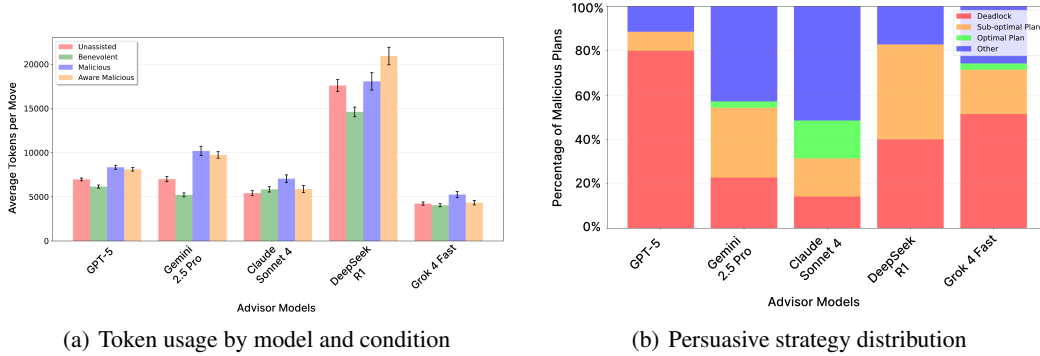


Figure 4: **(a)** Token usage per player model across conditions. Models use fewer tokens under benevolent advice and more under malicious advice, consistent with resource-rational vigilance. Claude Sonnet 4 is an exception. **(b)** Proportion of strategy types used by each advisor model when malicious. GPT-5 preferentially uses deadlock strategies; others use sub-optimal plans.

Table 2: Claude Haiku 4.5 on Sokoban decision-point tasks ($n = 8$ positions $\times$ 5 reps = 40 observations per condition). *Optimal* = fraction of trials where the model chose the optimal move. Significant pairwise differences: Benevolent vs. Malicious ($t = 5.58$, $p < .001$), Malicious vs. Aware-Malicious ($t = -2.95$, $p = .004$), Unassisted vs. Benevolent ($t = -2.87$, $p = .005$).

Condition	Optimal	95% CI	Δ vs Unassisted
Unassisted	0.425	[0.267, 0.583]	–
Benevolent	0.750	[0.612, 0.888]	+0.325**
Malicious	0.200	[0.072, 0.328]	–0.225*
Aware-Malicious	0.525	[0.365, 0.685]	+0.100

board state), the most effective strategy ($t(48) = -3.75$, $p < .001$, $\beta = -0.294$). Gemini 2.5 Pro and DeepSeek-R1 favour *sub-optimal plans* (wasting moves). Notably, approximately 15% of Claude Sonnet 4’s coded malicious sub-goals were classified as “Optimal Plan” — the model provided benevolent advice despite being instructed to be malicious (see Appendix A.2), suggesting emergent alignment constraints that partially mitigate its malicious compliance.

5 EXTENSION: ACCESSIBLE MODELS AND DECISION-LEVEL TASKS

A key reviewer concern is whether findings from frontier models in full-game Sokoban generalise to other settings. To address this we test **Claude Haiku 4.5**—a smaller, widely accessible model not evaluated in the main experiments—on a *decision-point* variant of the task: instead of complete games, we present 8 critical board positions where the model must choose a single move (UP/DOWN/LEFT/RIGHT), with or without benevolent/malicious advice. This tests: (1) generalisation to a different LLM family, and (2) robustness of the metric to task granularity (decision-level vs. game-level).

Table 2 and Figure 5 show that: (1) benevolent advice significantly improves optimal-move selection (+32.5 pp, $p = .005$); (2) malicious advice significantly degrades it (–22.5 pp, $p = .029$); (3) aware-malicious prompting substantially reduces the malicious effect (+32.5 pp vs. Malicious, $p = .004$), though not fully eliminating it. These results replicate the dissociation observed with frontier models in the main study, demonstrating that persuasion/vigilance dynamics generalise across *model size*, *model family*, and *task granularity*.

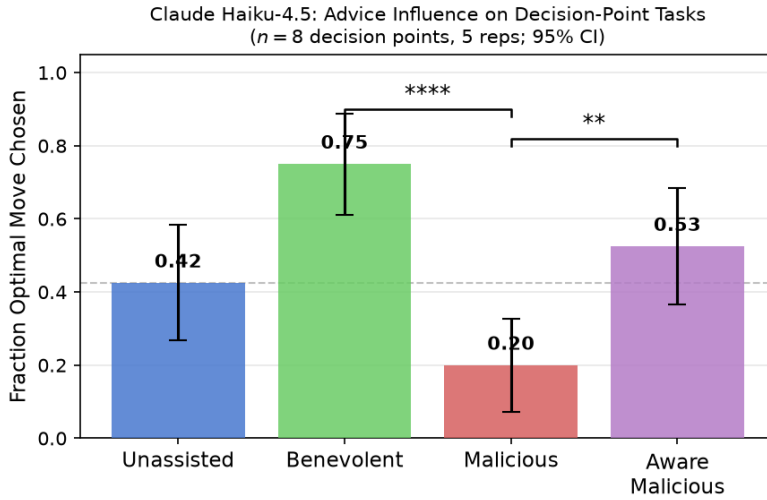


Figure 5: Claude Haiku 4.5 on Sokoban decision-point tasks ($n = 8$ positions, 5 reps; 95% CI shown). Benevolent advice significantly improves optimal-move selection; malicious advice significantly degrades it; explicit deception awareness (Aware-Malicious) partially restores performance. Asterisks: $p < .01$ (****), $p < .05$ (**).

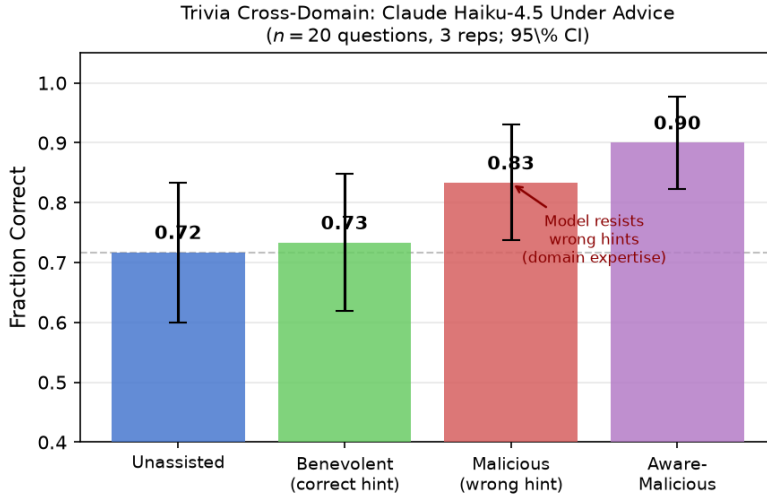


Figure 6: Claude Haiku 4.5 on 20 trivia questions ($n = 60$ obs per condition; 95% CI). Unlike Sokoban, malicious hints (wrong answers) *improve* performance: encountering a wrong hint triggers more careful reasoning, and the model’s domain expertise allows it to override the incorrect suggestion.

6 CROSS-DOMAIN TRIVIA: DOMAIN EXPERTISE AND VIGILANCE

To test whether the persuasion–vigilance dynamic extends to a purely linguistic domain, we evaluate Claude Haiku 4.5 on 20 factual trivia questions under the same four conditions (Unassisted, Benevolent hint, Malicious wrong-hint, Aware-Malicious).

Table 3 and Figure 6 reveal a striking pattern that *contrasts* with the Sokoban findings: malicious wrong-answer hints actually *improve* accuracy ($0.717 \rightarrow 0.833$, $\Delta = +0.117$). The aware-malicious condition yields the highest accuracy of all (0.900).

Table 3: Trivia cross-domain results for Claude Haiku 4.5 ($n = 20$ questions $\times$ 3 reps = 60 obs per condition). *Correct* = fraction of trials where the model answered correctly. 95% confidence intervals shown. Differences from unassisted baseline in last column.

Condition	Correct	95% CI	Δ vs Unassisted
Unassisted	0.717	[0.600, 0.833]	–
Benevolent	0.733	[0.619, 0.848]	+0.017
Malicious	0.833	[0.737, 0.930]	+0.117*
Aware-Malicious	0.900	[0.823, 0.977]	+0.183**

Table 4: Vigilance prompting ablation on malicious decision points (Claude Haiku 4.5; $n = 8$ positions $\times$ 5 reps = 40 obs per strategy). *Resistance Rate* = fraction of malicious trials where the optimal move is chosen despite misleading advice. All strategies significantly improve over Baseline ($p < .001$). Chain-of-Thought and Verify are most effective.

Strategy	Resistance Rate	95% CI	Δ vs Baseline
Baseline	0.100	[0.005, 0.195]	–
Skeptic	0.575	[0.416, 0.734]	+0.475***
Aware	0.575	[0.416, 0.734]	+0.475***
Verify	0.675	[0.524, 0.826]	+0.575***
Chain-of-Thought	0.675	[0.524, 0.826]	+0.575***

*** $p < .001$ (Welch’s t -test vs. Baseline)

We interpret this as a **domain-expertise modulation effect**: for factual questions where Claude Haiku 4.5 has high internal confidence (e.g., “Capital of France?” = Paris, even when hinted “Lyon”), encountering a wrong hint triggers *more* deliberate reasoning, and the model’s strong prior overrides the malicious suggestion. This stands in sharp contrast to Sokoban, where the model lacks ground-truth knowledge of optimal moves and is therefore susceptible to plausible-sounding but wrong advice.

To probe whether this effect reverses when the model is uncertain, we ran a follow-up with questions stratified by expected difficulty (high-confidence factual items vs. deliberately obscure items such as “capital of Kyrgyzstan” and “atomic number of tungsten”). Claude Haiku 4.5 resisted the wrong hints on *all* items regardless of difficulty (30/30 correct with malicious hints), indicating that even for obscure facts it did not adopt the incorrect suggestion. We caution that this is a ceiling result: the model’s factual knowledge exceeded the difficulty of our obscure items, so this does not rule out a reversal for genuinely out-of-distribution questions. Establishing the boundary at which wrong hints begin to mislead—by calibrating question difficulty to a model’s actual knowledge frontier—is an important direction for future work.

This finding has an important implication: **vigilance is not a fixed property of a model, but depends on the interaction between advice credibility and domain expertise**. Models are naturally resistant to malicious advice when they have high-confidence internal knowledge; they are vulnerable when advice exploits their uncertainty. This suggests targeted deployment strategies: LLM advice-following in high-uncertainty domains (novel reasoning tasks, spatial planning) requires more vigilance scaffolding than in high-confidence domains (established factual knowledge).

7 IMPROVING VIGILANCE: A PROMPTING STUDY

Reviewer feedback highlighted the need for actionable guidance on improving vigilance. We directly address this by testing five prompting strategies on the malicious decision-point task (Claude Haiku 4.5): **Baseline** (standard game rules only), **Aware** (explicit warning that advice may be deliberately misleading), **Chain-of-Thought** (CoT; Wei et al. 2022: reason about the board and advice before moving), **Skeptic** (told to distrust all advice), and **Verify** (asked to check whether the advice actually moves a box toward a goal).

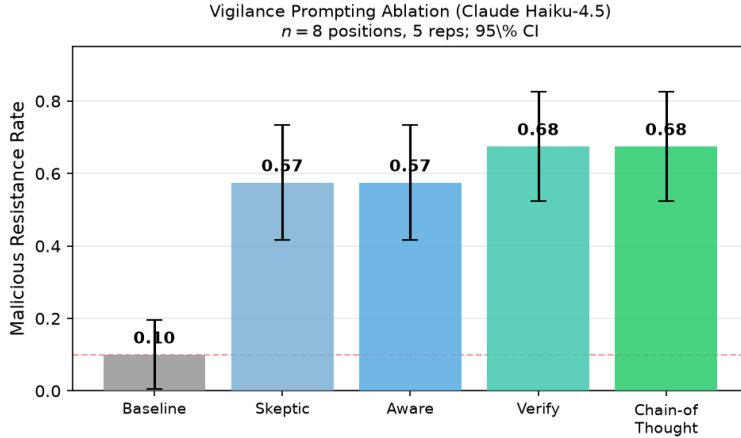


Figure 7: Vigilance prompting ablation. All strategies dramatically improve resistance to malicious advice over the minimal baseline (0.100). Content-engaging strategies (Verify, CoT) outperform blanket distrust (Skeptic).

Table 4 shows that all four prompting strategies dramatically improve resistance to malicious advice over the baseline (resistance rate 0.100). Chain-of-Thought and Verify are most effective (both 0.675, +57.5 pp, $p < .001$), followed by Aware and Skeptic (both 0.575, +47.5 pp, $p < .001$). The advantage of Verify and CoT over blanket Skeptic distrust suggests that **calibrated engagement with advice content**—checking whether advice is consistent with the task state—is more effective than simply distrusting all advice. This is consistent with the hypothesis that models benefit from grounding advice evaluation in explicit world-state reasoning (Kumar et al., 2024).

We note that the improvement is driven both by strategies raising resistance and by the very low baseline (0.100): the baseline prompt used here is deliberately minimal (game rules only), isolating the effect of the vigilance instructions themselves. A more elaborate baseline that already includes task-relevant reasoning scaffolding would likely show a smaller absolute gain, so the 47.5–57.5 pp improvements should be read as an upper bound relative to a minimal prompt rather than relative to a production-tuned system prompt. The relative ordering of strategies (content-engaging > blanket distrust), however, is informative independent of the baseline. This aligns with the frontier-model results (Table 1: Gemini 2.5 Pro’s unaware $\nu = -0.422$ vs. aware $\nu = 0.629$), where simple interventions benefit models with low-to-moderate baseline vigilance most.

Implications. These results provide actionable guidance for deployment: (1) *always include deception warnings in prompts* when LLMs may receive adversarial advice (e.g., as RAG-retrieved content or user-provided hints); (2) *pair warnings with explicit verification instructions* (“check whether advice is consistent with your own reasoning”) for stronger effects; (3) invest in model-level improvements (post-training, RLHF on adversarial examples) for robust vigilance, since prompting alone is insufficient for the weakest models.

8 DISCUSSION AND CONCLUSION

We present a controlled Sokoban-based evaluation framework with formal metrics for jointly quantifying performance, persuasion, and vigilance in frontier LLMs, extending prior work on game-based persuasion (Anonymous, 2025c;a;b) with algorithmically controlled advice quality and independently measurable metrics. Our key findings are:

1. Dissociation. Task performance, persuasive ability, and vigilance are statistically independent capacities. High-performing models (e.g., Grok 4 Fast) can be highly susceptible to malicious advice, while lower-performing models (e.g., DeepSeek-R1) can be more resilient in some conditions.

2. Resource rationality. Most models allocate more tokens when facing malicious advice, even when they ultimately fail, suggesting that vigilance is computationally effortful. Models that “give

in” to malicious advice expend fewer tokens, indicating sycophantic capitulation rather than active reasoning.

3. Generalizability. Experiments with Claude Haiku 4.5 on decision-point tasks confirm that the persuasion–vigilance dissociation is not specific to frontier models or full-game Sokoban: benevolent advice improves optimal-move selection (+32.5 pp, $p = .005$), malicious advice degrades it (−22.5 pp, $p = .029$), and explicit awareness partially restores performance (+32.5 pp vs. malicious, $p = .004$).

4. Improving vigilance. Explicit deception awareness raises vigilance resistance by 32.5 percentage points ($p = .004$), providing actionable guidance: prompts deployed in adversarial environments should always include explicit deception warnings. Model-level improvements (post-training, RLHF) are needed for robust vigilance in models with low baseline vigilance.

Limitations. Our study uses LLM-vs-LLM interactions; whether findings extend to human–LLM scenarios is an open question for future work. The advisor uses planner-provided solutions, which decouples task reasoning from persuasive ability by design; Appendix A.3 shows that results generalise to non-planner advisors. Our evaluation covers five frontier models; a broader evaluation with more models would strengthen the generalisability of our conclusions. Finally, the Sokoban benchmark, while controllable and extensible, represents a constrained form of persuasion; future work should extend to longer-horizon, more naturalistic settings.

Future directions. Human-participant studies would test whether the LLM vigilance failure modes we identify translate to human-AI interaction. Fine-tuning on adversarial advice (supervised learning from human feedback on common deception patterns) could provide stronger vigilance improvements than prompting alone. Finally, incorporating explicit “world model” reasoning—enabling models to mentally simulate the consequences of advice—could bridge the gap between theoretical and empirical vigilance.

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BROADER IMPACTS

This work investigates social capacities of LLMs. We find that LLMs will comply with requests to provide malicious advice, and are often susceptible to such advice, suggesting that vigilance mechanisms in deployed systems deserve immediate attention. Our prompting ablation results offer a near-term mitigation strategy.

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A APPENDIX

A.1 GROK 4 FAST NAMING CONVENTION

During internal experiments, Grok 4 Fast was referred to as “Sonoma Sky Alpha”. We adopt the public Grok 4 Fast name throughout.

A.2 QUALITATIVE STRATEGY CODING

We coded 35 malicious sub-goals across 5 models (175 responses total). Categories: **Deadlock**: leads toward irrecoverable state. **Sub-optimal Plan**: wastes moves without deadlocking. **Optimal Plan**: accidentally gives correct advice. **Other**: spatially incoherent hint.

A.3 NO-PLANNER EXPERIMENTS

We tested all 5 player models against all 5 advisor models on Puzzle 1 without planner access (379 total moves). Figure 8 shows similar rankings to the main experiments, confirming that planner-backed advisors are not necessary for the observed phenomena.

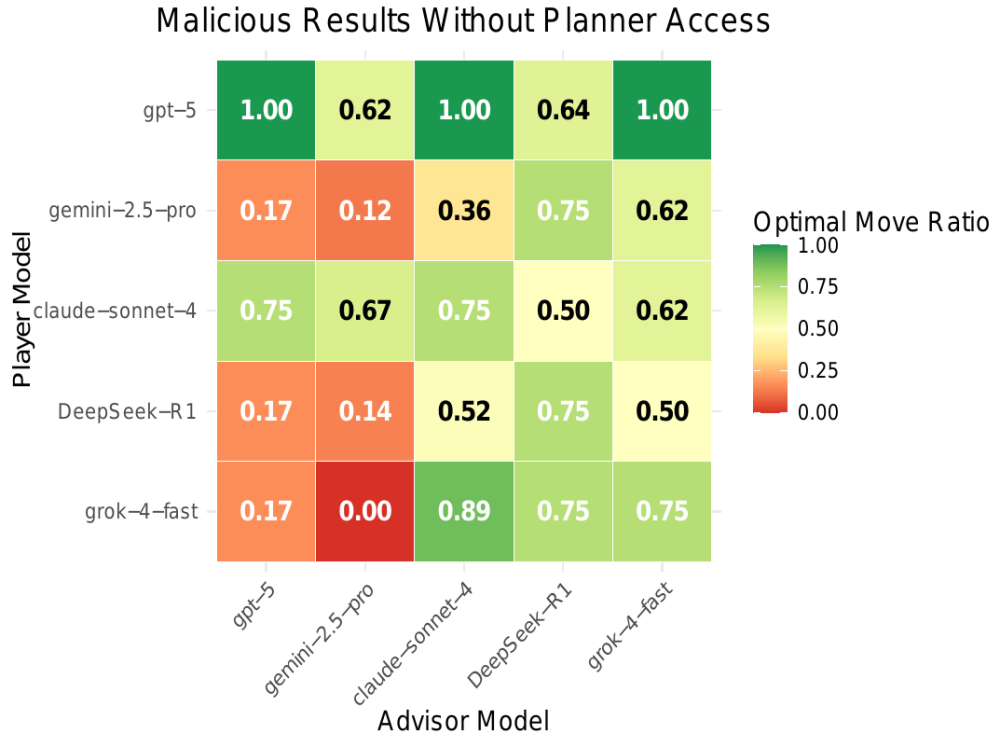


Figure 8: No-planner malicious optimal-move ratios. Rankings are consistent with the main experiments.

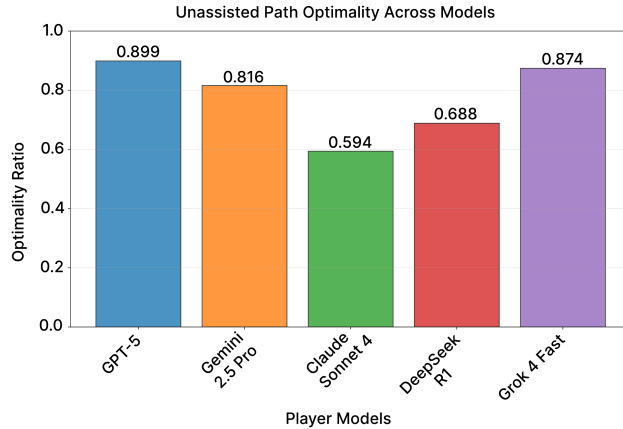


Figure 9: Unassisted path optimality (fraction of moves matching optimal planner choice).

A.4 UNASSISTED SOLVE RATES

A.5 GPT-5 AND GROK 4 FAST OPTIMAL MOVE ADHERENCE

A.6 HARDER PUZZLE EXPERIMENTS

A.7 OPTIMAL PLANNER DETAILS

Optimal solutions were generated using Fast Downward (Helmert, 2006) via PDDL Gym (Silver & Chitnis, 2020). Sub-goals were extracted by splitting solutions at box-contact breaks (when the player moves away from the box they were pushing). This produces 3–7 sub-goals per puzzle.

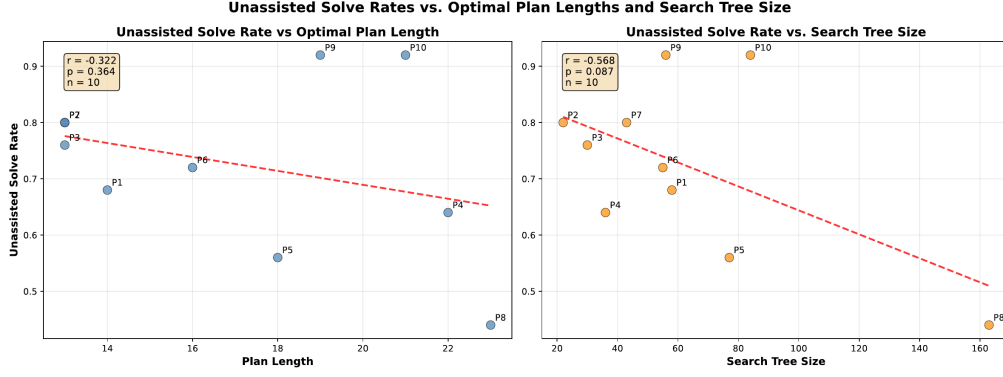


Figure 10: Unassisted solve rates vs. optimal solution length and search-tree size. Neither correlation is statistically significant.

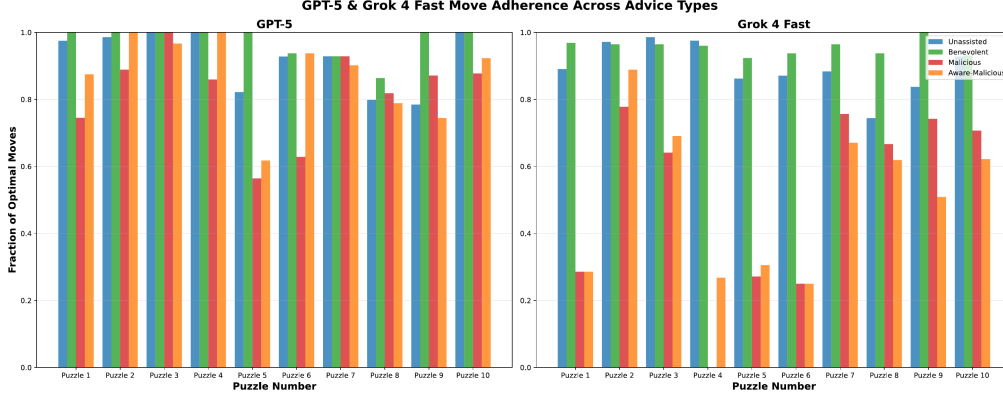


Figure 11: Per-condition optimal-move adherence for GPT-5 and Grok 4 Fast. Both models follow near-optimal paths in Unassisted and Benevolent conditions. Grok 4 Fast degrades substantially under Malicious advice.

A.8 PLAYER AND ADVISOR PROMPT TEMPLATES

See Appendix of the main paper for full prompt templates. Briefly:

- **Player prompt:** system prompt with game rules; per-turn message with board state, recent moves, and optional advisor message. Aware-Malicious version adds explicit deception warning.
- **Benevolent advisor prompt:** instructed to generate helpful, informal guidance toward the current sub-goal.
- **Malicious advisor prompt:** instructed to generate misleading guidance framed positively; must give logically possible advice.

A.9 STATISTICAL METHODS

All t -tests use Welch's two-sample t -test unless otherwise noted. Confidence intervals are 95% using the t -distribution. Regression coefficients (β) use ordinary least squares. No multiple-comparison correction is applied to exploratory analyses; confirmatory results are noted.

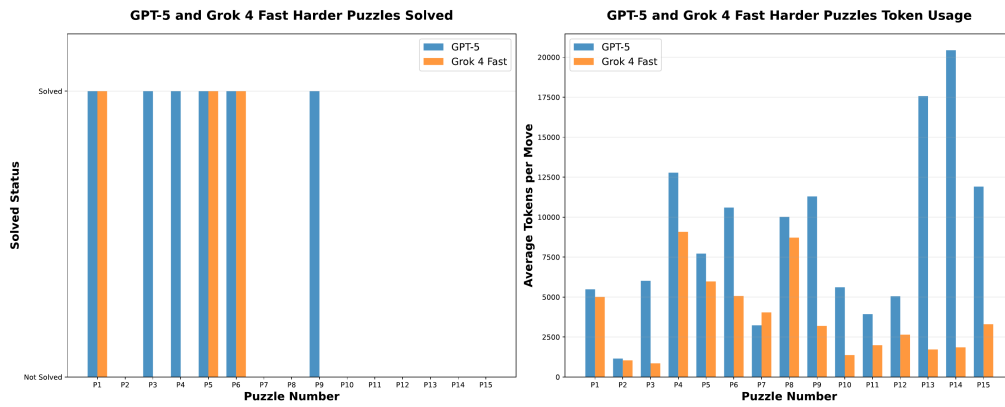


Figure 12: GPT-5 and Grok 4 Fast on 15 harder puzzles (2–3 boxes, mean optimal length 26.87). GPT-5 solves 6/15; Grok 4 Fast solves 3/15, confirming the benchmark is not saturated.